01 / PROBLEM AND CONTRIBUTION

Asynchronous FIFO & alternating streams

Advanced Digital Design & Verification · Lab 1 | Fall 2026 | Two-person team

RESULT AT A GLANCE

Verified a 16-entry asynchronous FIFO and two 40-entry alternating-stream queues, correcting parity advancement so the reader waits for the required queue.

The engineering question

How can data order and boundary behavior be checked when reads and writes occur in different clock domains?

Situation

Different read and write clocks stress FIFO boundary behavior. A second circuit combines even and odd streams, where a missing required entry must not advance the output sequence.

Task

Verify flags and ordering, choose queue depth from the tested traffic pattern, and correct the alternating read-state behavior.

My contribution

Developed tests and flag checks for the asynchronous FIFO; worked on FIFO modules, input routing, sizing, and output-order verification for the alternating circuit.

Tools and concepts

SystemVerilog / Clock-domain crossing / FIFO design / VCS / Verdi

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Verified fill/full/drain behavior, almost-full and almost-empty thresholds, and simultaneous reads/writes with different clock rates.

02 / Implementation step

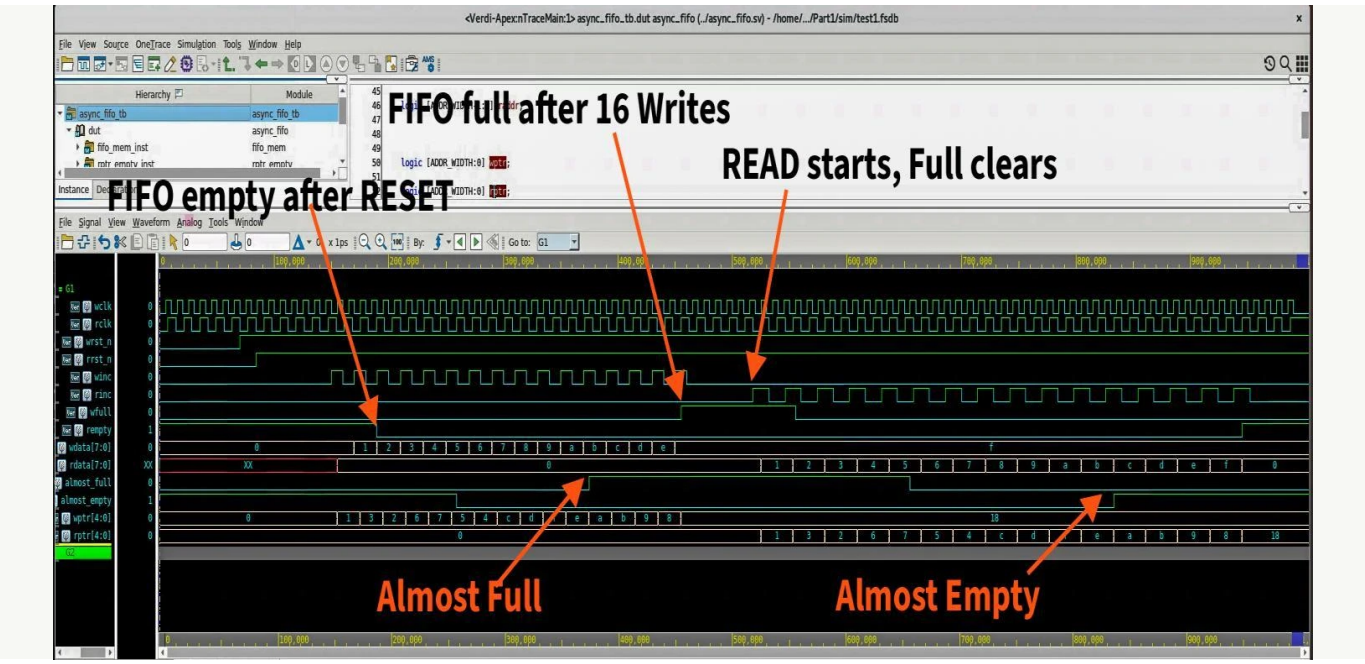
Added internal occupancy signals to the waveform to locate status transitions: almost-full at 12 entries and almost-empty at 4 entries in the 16-entry FIFO.

03 / Implementation step

For the even/odd stream, used two 40-entry FIFOs. Corrected the read controller so parity advances only after a valid read; it waits when the required queue is empty.

Boundary conditions

The 16-entry FIFO tests cover fill, drain, status thresholds, and simultaneous traffic. The almost-full and almost-empty observations are tied to occupancies of 12 and 4 entries in the documented setup.



FIFO fill and read sequence

Annotated simulation: full after 16 writes, followed by reads.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

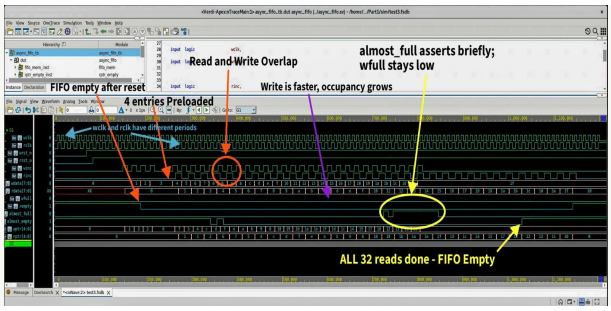
Asynchronous FIFO	16 entries
Alternating queues	40 entries each
Status thresholds	12 / 4 entries

Reported result

The two alternating-stream tests reached peak occupancies of 40/24 and 32/31 entries in the even/odd FIFOs. Waveforms confirmed output ordering after the read-state correction.

Alternation is a control requirement

In the two-queue circuit, the reader must wait if the next required even or odd queue is empty. Advancing parity without accepting a valid read would break ordering even if each FIFO individually stored its data correctly.



Overlapping reads and writes

Reported asynchronous FIFO waveform with occupancy annotations.

Scope and limitations

The depth of 40 is justified by the tested arrival pattern. CDC simulation and synthesis are not a substitute for a dedicated CDC sign-off flow.

Possible next steps - proposed, not completed results

- Extend clock-ratio and burst-pattern testing beyond the two demonstrated occupancy patterns.
- Run an appropriate dedicated CDC review before making clock-domain crossing sign-off claims.

Source material: Team Lab 1 report, pp. 1-10.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.